Enhancement of immunity in the elderly by dietary supplementation with the probiotic Bifidobacterium lactis HN019.

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BACKGROUND: The aging process can lead to a decline in cellular immunity. Therefore, the elderly could benefit from safe and effective interventions that restore cellular immune functions. OBJECTIVE: We determined whether dietary supplementation with the known immunostimulating probiotic Bifidobacterium lactis HN019 could enhance aspects of cellular immunity in elderly subjects. DESIGN: Thirty healthy elderly volunteers (age range: 63-84 y; median: 69 y) participated in a 3-stage dietary supplementation trial lasting 9 wk. During stage 1 (run-in), subjects consumed low-fat milk (200 mL twice daily for 3 wk) as a base-diet control. During stage 2 (intervention), they consumed milk supplemented with B. lactis HN019 in a typical dose (5 x 10(10) organisms/d) or a low dose (5 x 10(9) organisms/d) for 3 wk. During stage 3 (washout), they consumed low-fat milk for 3 wk. Changes in the relative proportions of leukocyte subsets and ex vivo leukocyte phagocytic and tumor-cell-killing activity were determined longitudinally by assaying peripheral blood samples. RESULTS: Increases in the proportions of total, helper (CD4(+)), and activated (CD25(+)) T lymphocytes and natural killer cells were measured in the subjects' blood after consumption of B. lactis HN019. The ex vivo phagocytic capacity of mononuclear and polymorphonuclear phagocytes and the tumoricidal activity of natural killer cells were also elevated after B. lactis HN019 consumption. The greatest changes in immunity were found in subjects who had poor pretreatment immune responses. In general, the 2 doses of B. lactis HN019 had similar effectiveness. CONCLUSION: B. lactis HN019 could be an effective probiotic dietary supplement for enhancing some aspects of cellular immunity in the elderly.